

Pointing Error Effects on Free-Space Optical Communication Links in the Presence of Atmospheric Turbulence

Deva K. Borah, *Member, IEEE*, and David G. Voelz

Abstract—The effects of pointing errors on the performance of a free-space optical laser communication system are studied. The log-normal and gamma-gamma irradiance probability density function models have been considered to include the effects of optical turbulence. An expression for the mean intensity under pointing fluctuations is derived. Results for ergodic and outage channel capacities are presented. Since the irradiance statistics vary with pointing fluctuations, a wave-optics-based approach is proposed to evaluate the capacity expressions. Although, in general, turbulence degrades communications link performance, it can reduce sensitivity to pointing errors by increasing the effective beam radius. Finally, we present analytical expressions for evaluating the bit-error-rate performance of the *on-off* keying system in the presence of pointing errors, and provide error performance results.

Index Terms—Atmospheric turbulence, digital communication, information rates, intensity modulation, pointing errors.

I. INTRODUCTION

FREE-SPACE OPTICAL (FSO) communication channels provide several advantages over radio frequency (RF) channels. They offer high bandwidth, secure transmission, and more freedom from interference. Further, these links can be established with components of smaller size and weight. However, there are two major issues with FSO links. First, they are affected by the atmospheric channel conditions such as scattering, absorption, and turbulence. Second, FSO links require accurate pointing.

The atmospheric channel conditions such as scattering and absorption mainly cause power loss in the received signal. For optical links covering a few hundred meters or more distances, atmospheric turbulence can cause severe error rates due to the random fluctuations in the received signal. There have been several critical studies on the effects of atmospheric turbulence on the error rates of FSO links. A major focus has been on *on-off* keying (OOK) systems since higher order modulations are typically too complex for practical implementation. The authors in

[1] study maximum likelihood (ML) symbol by symbol detection techniques for log-normal (LN) turbulence channels. The fading state is considered unknown but the channel statistics are assumed known. For known temporal correlation, ML sequence detection is also studied. The performance of block, convolutional, and turbo codes have been studied for the weak turbulence regimes by the same authors in [2].

The channel capacity of FSO links has been investigated by several authors. The capacity for the OOK optical channel under LN fading has been presented in [3] for known channel state information. The capacity is shown to decrease with an increase in turbulence strength. For known channel statistics, ergodic capacity results are given in [4] for the gamma-gamma model. For weak turbulence, characterized by a low Rytov variance σ_R^2 , the capacity is found to increase quickly with an increase in the signal-to-noise ratio (SNR) in contrast to the high σ_R^2 region where the increase is rather sluggish. In general, for a given SNR, the capacity at low σ_R^2 is found to be higher than the capacity at higher σ_R^2 , except in a saturation region, i.e., large σ_R^2 , where this monotonic behavior disappears.

In addition to the effects of atmospheric turbulence, FSO links are also highly dependent on the pointing performance. The pointing errors can arise due to mechanical misalignment, errors in tracking systems, or due to mechanical vibrations present in the system. Pointing errors can be thought of being composed of two components: a fixed error, called *boresight*, and a random error, called *jitter*. The behaviors of pointing errors and related issues are treated in [5], where the irradiance probability density function (pdf) due to boresight and jitter has been derived. Some other related pointing error discussions are available in [6]–[9].

Since optical systems require precise pointing, the effect of pointing errors on link performance is a vital issue. A bit-error-rate (BER) model was developed in [10] to take into account pointing errors due to building sways in the presence of atmospheric turbulence. A BER analysis for the K distributed turbulence model in the presence of pointing errors was also presented in [11]. Recently, an analysis on the outage capacity with jitter errors was presented for LN and gamma-gamma distributed turbulence models in [12]. However, we are not aware of any method that demonstrates how to incorporate the variations of the irradiance pdfs over the receiver plane into capacity calculations. For example, the irradiance pdf at the beam center is different from the irradiance pdf at an off axis position. Thus, as the beam undergoes random pointing fluctuations, the variations of the received sample statistics must be taken into account in designing and simulating such links.

Manuscript received August 06, 2008; revised January 16, 2009 and April 15, 2009. First published May 12, 2009; current version published August 21, 2009. This work was supported in part by the National Science Foundation under Grant ECS-0636512 and in part by the Air Force Office of Scientific Research (AFOSR) under Grant FA9550-04-1-0392.

The authors are with the Department of Electrical and Computer Engineering, Klipsch School of Electrical Computer Engineering, New Mexico State University, Las Cruces, NM 88003 USA (e-mail: dborah@nmsu.edu; davvoelz@nmsu.edu).

Digital Object Identifier 10.1109/JLT.2009.2022771

The purpose of this paper is twofold. First, we investigate the effect of beam pointing errors on the capacity of optical links affected by atmospheric turbulence. We derive both ergodic and outage capacity expressions. Second, we provide a simulation and analysis technique for designers studying such systems. A wave-optics-based approach is proposed to handle the irradiance pdf variations over the receiver plane for evaluating the capacity expressions. Our results show that turbulence can be helpful for ergodic capacity when large pointing errors are present in the system. A similar benefit in the outage capacity is not observed as the improvement in the pdf characteristics in the low-signal region is not large enough at the low outage probability. In the presence of boresight errors, a nonzero jitter can increase the average received intensity, and we provide an expression for the jitter value that maximizes the average received intensity. Finally, analytical BER expressions for uncoded systems are presented and error performance results are demonstrated.

The paper is organized as follows. Section II gives the system model. In Section III, LN and gamma-gamma turbulence models in the presence of pointing errors are described. Capacity expressions are derived in Section IV, and a wave-optics-based strategy to evaluate the capacity expressions is presented. This section also demonstrates the benefit of turbulence in the presence of pointing errors. Section V provides analytical BER expressions for uncoded systems. The error performance results are also demonstrated. Finally, Section VI presents conclusions of our study.

II. SYSTEM MODEL

Consider an intensity modulation with direct detection (IM/DD) link that uses OOK. The transmitter uses the lowest order transverse electromagnetic Gaussian beam. After propagation through the atmospheric turbulence channel, the mean irradiance profile $\bar{I}(r)$ on the receiver plane is modeled as [13]

$$\bar{I}(r) = K_T \exp\left(-\frac{r^2}{2\Omega_T^2}\right) \quad (1)$$

where r is the distance from the mean beam center on the transverse plane, $K_T = W_0^2/W_e^2$ is the gain constant, W_0 is the effective beam radius at the transmitter, $W_e = W(1 + 1.33C_n^2\Lambda^{5/6})^{1/2}$ is the effective beam spot radius at the receiver, W is the receiver beam radius in free space, Λ equals $2L/kW^2$, L is the distance between the transmitter and the receiver, k is the optical wavenumber, and $\Omega_T = W_e/2$. We assume plane wave at the source. The Rytov variance is given by $\sigma_R^2 = 1.23C_n^2k^{7/6}L^{11/6}$, where C_n^2 is the index of refraction structure parameter of the atmosphere expressed in units of $\text{m}^{-2/3}$. Ideally, one would like to keep the receiver aperture fixed on the mean beam peak so that $r = 0$, but due to pointing difficulties, r is a random variable.

Our study mostly considers a point detector. For the weak turbulence regimes, the results so obtained are directly applicable for detector aperture diameters much smaller than the turbulence coherence diameter [13]. For larger apertures, the capacity results serve as the lower bound and the error analysis results as the upper bound. The method presented in this paper, however, can be used to study the effects of larger apertures as well.

The pointing error consists of two components: a fixed error, called *boresight* and denoted by A , and a random error, called *jitter*, superimposed over the fixed boresight error. Due to boresight and jitter, each received intensity sample can be thought of as a randomly sampled point on a random irradiance profile. Jitter is modeled as a 2-D random variable with independent components u and v so that the random sampling point has co-ordinates $(A + u, v)$, and thus, $r = \sqrt{(A + u)^2 + v^2}$. We also define $\rho = r^2 = (A + u)^2 + v^2$. Since u and v are independent, the pdf of ρ can be obtained by convolving the pdfs of v^2 and $(A + u)^2$. For Gaussian u and v , the pdf of ρ becomes Ricean as [5]

$$p(\rho) = \begin{cases} \frac{1}{2\sigma_j^2} \exp\left(-\frac{1}{2\sigma_j^2}(\rho + A^2)\right) I_0\left(\frac{A}{\sigma_j^2}\sqrt{\rho}\right), & \sigma_j \neq 0 \\ \delta(\rho - A^2), & \sigma_j = 0 \end{cases} \quad (2)$$

where σ_j is the jitter standard deviation, $I_0(\cdot)$ is the modified Bessel function of the first kind and order zero, and $\delta(\cdot)$ is the unit impulse function. We refer σ_j simply as the jitter. For a given ρ , if the received signal flux is $I_s(\rho)$, then the electrical signal is $\eta I_s(\rho)$, where η is the optical-to-electrical conversion efficiency [1]. The receiver is assumed to integrate the signal over each bit period. For simplicity, we keep $\eta = 1$, and also sometimes simply write I_s instead of $I_s(\rho)$. The received electrical signal by a point receiver can, therefore, be modeled as

$$y = I_s x + n \quad (3)$$

where $x \in \{0, 1\}$ is the binary data symbol, n is zero-mean additive white Gaussian noise (AWGN) of variance $\sigma_n^2 = N_0/2$, and $N_0/2$ is the two-sided noise power spectral density [1], [4]. Note that I_s is a normalized signal, and it is normalized such that $E[I_s(0)] = 1$ at the mean peak, i.e., without any boresight and jitter. The notation $E[\cdot]$ denotes expectation. This normalization enables us to investigate performance loss due to pointing errors. Note this normalization also implies that the reference irradiance level is at the receiver plane, and so, the transmitter irradiance levels are, in general, different for different turbulence and pointing conditions. The quantity I_s is obtained from the unnormalized value I by $I_s = \gamma I$, where $\gamma = W_e^2/W_0^2$ is a normalization constant. The SNR is defined as $\eta^2 E[I_s]^2/N_0$, and it becomes $E[I_s]^2/N_0$ for $\eta = 1$.

Considering the effects of pointing errors, the pdf of I_s is written as

$$p(I_s) = \int_0^\infty p(I_s|\rho)p(\rho)d\rho \quad (4)$$

where $p(I_s|\rho)$ denotes conditional pdf. Two pdf models for $p(I_s|\rho)$ will be discussed in Section III. Note that for a given value of ρ , the random variable $I_s(\rho)$ has a local mean of $\bar{I}_s(\rho) = \gamma \bar{I}(r)$, where $\bar{I}(r)$ is given by (1).

We now derive the global mean $\langle I_s \rangle$ over all ρ values for a given boresight A as follows:

$$\begin{aligned} \langle I_s \rangle &= \int_0^\infty \int_0^\infty I_s p(I_s|\rho) \frac{1}{2\sigma_j^2} \\ &\quad \times \exp\left(-\frac{1}{2\sigma_j^2}(\rho + A^2)\right) I_0\left(\frac{A}{\sigma_j^2}\sqrt{\rho}\right) dI_s d\rho \end{aligned}$$

$$\begin{aligned}
&= \gamma \int_0^\infty \bar{I}(\sqrt{\rho}) \frac{1}{2\sigma_j^2} \exp\left(-\frac{1}{2\sigma_j^2}(\rho + A^2)\right) \\
&\quad \times I_0\left(\frac{A}{\sigma_j^2}\sqrt{\rho}\right) d\rho \\
&= \gamma \frac{1}{2\sigma_j^2} \frac{W_0^2}{W_e^2} \exp\left(-\frac{A^2}{2\sigma_j^2}\right) \\
&\quad \times \int_0^\infty \exp\left(-\left(\frac{2}{W_e^2} + \frac{1}{2\sigma_j^2}\right)\rho\right) I_0\left(\frac{A}{\sigma_j^2}\sqrt{\rho}\right) d\rho \\
&= \gamma \frac{W_0^2}{W_e^2 + 4\sigma_j^2} \exp\left(-\frac{2A^2}{W_e^2 + 4\sigma_j^2}\right) \quad (5)
\end{aligned}$$

where $\int_0^\infty \exp(-\alpha x) I_0(2\sqrt{\beta x}) dx = 1/\alpha \exp(\beta/\alpha)$ [14] is used. Note that the overall mean value $\langle I_s \rangle$ does not depend on a specific turbulence pdf model.

III. ATMOSPHERIC TURBULENCE MODELS

A. Log-Normal Model

Under weak turbulence scenarios, the LN distribution has been found to be a good model for I_s although it can underestimate the peak irradiance value and the distribution wings as compared with experimental measurements [13], [15]. Under this model, we write

$$\begin{aligned}
p(I_s|\rho) &= \frac{1}{I_s \sigma(\rho) \sqrt{2\pi}} \\
&\quad \times \exp\left(-\frac{\left(\ln\left(\frac{I_s}{\gamma \bar{I}(\sqrt{\rho})}\right) + \frac{1}{2}\sigma^2(\rho)\right)^2}{2\sigma^2(\rho)}\right) \quad (6)
\end{aligned}$$

where $\sigma^2(\rho)$, the log-irradiance variance, is the parameter of the LN model.

B. Gamma-Gamma PDF Model

The gamma-gamma pdf model was a subsequent development to the LN model and is a two-parameter distribution that assumes small-scale fluctuations are modulated by large-scale fluctuations. In this case, $p(I_s|\rho)$ is given by

$$\begin{aligned}
p(I_s|\rho) &= \frac{2(\alpha(\rho)\beta(\rho))^{(\alpha(\rho)+\beta(\rho))/2}}{\gamma \bar{I}(\sqrt{\rho}) \Gamma(\alpha(\rho)) \Gamma(\beta(\rho))} \\
&\quad \times \left(\frac{I_s}{\gamma \bar{I}(\sqrt{\rho})}\right)^{(\alpha(\rho)+\beta(\rho))/2-1} \\
&\quad \times K_{\alpha(\rho)-\beta(\rho)}\left(2\sqrt{\alpha(\rho)\beta(\rho)} I_s / (\gamma \bar{I}(\sqrt{\rho}))\right) \quad (7)
\end{aligned}$$

where $\Gamma(\cdot)$ is the gamma function, $\alpha(\rho)$ and $\beta(\rho)$ are model parameters that depend on the location in the receiver plane, and $K_{\alpha(\rho)-\beta(\rho)}(\cdot)$ is the modified Bessel function of the second kind of order $\alpha(\rho) - \beta(\rho)$.

The turbulence pdf model parameters, in general, vary with ρ . To our best knowledge, a theoretical model describing variations of α and β with ρ is still an open issue. In our study, we obtain parameter variations using a wave-optics-based approach to be described later.

IV. CAPACITY CALCULATION

Since the channel is time-varying due to atmospheric turbulence and pointing fluctuations, the information carrying capacity, in bits per channel use, of the channel is also time-varying. One way to characterize this channel is to consider an average information carrying rate through the channel with an arbitrarily small probability of detection error. This average rate or capacity can be approached by using channel codes of large block lengths so that the effects of the channel variations are captured well by the codeword. This is the ergodic capacity of the channel. For example, when channel fluctuations occur in the order of milliseconds [16], codewords of sufficient lengths, for example, as considered in [17] and [18], can be used to capture multiple fading cycles. In contrast, if the channel variations are relatively slow, then the channel or the received irradiance intensity can remain unchanged for a significantly long period of time. In this case, system outages will occur when the irradiance intensity falls below a threshold. Thus, an alternative capacity measure is the capacity under a specified outage condition or simply the outage capacity. This is the maximum rate of transmission so that the probability of the system being in an outage meets a specified value P_{out} . We consider both ergodic and outage capacities for turbulence channels with pointing errors. Further, our capacity results correspond to only binary OOK input symbols.

A. Ergodic Capacity for Known Channel at the Receiver

In this case, the receiver is assumed to know the channel or signal I_s for each received sample. We have to maximize the mutual information between input x and output y for given channel values. The mutual information becomes

$$\begin{aligned}
E &\left[\log_2 \left(\frac{p(y|x, I_s)}{\sum_{m=0,1} p(y|x = m, I_s) P_X(m)} \right) \right] \\
&= \sum_{x=0}^1 P_X(x) \int_{-\infty}^{\infty} \int_0^{\infty} p(y|x, I_s) p(I_s) \\
&\quad \times \log_2 \left(\frac{p(y|x, I_s)}{\sum_{m=0,1} p(y|x = m, I_s) P_X(m)} \right) dI_s dy \\
&= - \sum_{m=0}^1 \frac{P_X(m)}{\sqrt{2\pi\sigma_n^2}} \int_{-\infty}^{\infty} \int_0^{\infty} p(I_s) \exp\left(-\frac{(y - mI_s)^2}{2\sigma_n^2}\right) \\
&\quad \times \log_2 \left(P_X(m) + P_X(\bar{m}) \right) \\
&\quad \times \exp\left(\frac{(1-2m)(2yI_s - I_s^2)}{2\sigma_n^2}\right) dI_s dy \quad (8)
\end{aligned}$$

where $P_X(m)$ is the probability of the bit being one ($m = 1$) or zero ($m = 0$), and $\bar{m}$ is the complement of m . The pdf $p(I_s)$ in (8) is computed using (4). This expression is numerically evaluated. In the absence of atmospheric turbulence, the pdf $p(I_s)$ becomes (9), shown at the bottom of the next page, [5] for $I_s > 0$, where $\Omega = W/2$.

B. Ergodic Capacity for Unknown Channel at the Receiver

In this case, the receiver does not know the channel I_s for each sample. However, the pdfs $p(y|x = 0)$ and $p(y|x = 1)$

are assumed known. Thus, we have to maximize the following mutual information:

$$E \left[\log_2 \left(\frac{p(y|x)}{\sum_{m=0,1} p(y|x=m)P_X(m)} \right) \right] \\ = \sum_{x=0}^1 P_X(x) \int_{-\infty}^{\infty} p(y|x) \\ \times \log_2 \left(\frac{p(y|x)}{\sum_{m=0,1} p(y|x=m)P_X(m)} \right) dy \quad (10)$$

where $p(y|x)$ is evaluated as

$$p(y|x) = \int_0^{\infty} p(y|x, I_s) p(I_s) dI_s$$

and it takes the form

$$p(y|x) = \begin{cases} \frac{1}{\sqrt{2\pi\sigma_n^2}} \exp\left(-\frac{y^2}{2\sigma_n^2}\right), & x = 0 \\ \int_0^{\infty} \frac{1}{\sqrt{2\pi\sigma_n^2}} \exp\left(-\frac{(y-I_s)^2}{2\sigma_n^2}\right) p(I_s) dI_s, & x = 1. \end{cases}$$

The pdf $p(I_s)$ is obtained from (2) and (4). In the case of no atmospheric turbulence, the previous expression uses the pdf $p(I_s)$ given by (9).

C. Outage Capacity

The outage capacity is the largest transmission rate R_o for the given outage probability P_{out} obtained as

$$P_{\text{out}} = \text{Prob}(C(I_{so}, \bar{\xi}) < R_o)$$

where

$$C(I_{so}, \bar{\xi}) = \sum_{x=0}^1 P_X(x) \int_{-\infty}^{\infty} p(y|x, I_{so}) \\ \times \log_2 \left(\frac{p(y|x, I_{so})}{\sum_{m=0,1} p(y|x=m, I_{so})P_X(m)} \right) dy \\ = - \sum_{m=0}^1 \frac{P_X(m)}{\sqrt{2\pi\sigma_n^2}} \int_{-\infty}^{\infty} \exp\left(-\frac{(y-mI_{so})^2}{2\sigma_n^2}\right) \\ \times \log_2 \left(P_X(m) + P_X(\bar{m}) \right. \\ \left. \times \exp\left(\frac{(1-2m)(2yI_{so} - I_{so}^2)}{2\sigma_n^2}\right) \right) dy \quad (11)$$

where I_{so} is a threshold value and $\bar{\xi}$ is the average SNR over signal and noise values. For a given P_{out} , we can find I_{so} equiv-

alently from $P_{\text{out}} = \text{Prob}(I_s \leq I_{so}) = F_s(I_{so})$, where $F_s(\cdot)$ is the cumulative distribution function (CDF) of I_s .

The cause for capacity variations due to turbulence and pointing errors can be divided into two factors by writing the received signal as $I_s = \gamma_s I_\nu$, where I_ν is the normalized signal so that $E[I_\nu] = 1$, and γ_s is a normalization constant. When the turbulence conditions change, say from weaker turbulence to stronger turbulence regimes, and/or pointing errors occur, both the value of γ_s and the pdf of I_ν change too. For example, when the center of the beam drifts away from the receiver, the value of γ_s decreases, and at the same time, the pdf of I_ν changes to represent relatively larger fluctuations. The effect of γ_s can be compensated by increasing the transmit irradiance intensity. An important question is: how much further intensity increase is required to compensate for the new pdf of I_ν ? Observe from (11) that for a given noise level, capacity is affected by I_{so} . Recalling $I_s = \gamma_s I_\nu$, we find $I_{so} = \gamma_s F_\nu^{-1}(P_{\text{out}})$, where $F_\nu(\cdot)$ is the CDF of I_ν . Thus, the new pdf of I_ν affects capacity through the term $F_\nu^{-1}(P_{\text{out}})$. For example, for the system parameters to be described later for the weak turbulence regime, we find that for a boresight of 24 cm, an SNR increase of 11.46 dB is required to get the same outage capacity as at the beam center. This SNR consists of two parts: 9.69 dB to compensate for the effect of γ_s and 1.77 dB to compensate for the effects of the new pdf. Similarly, if the turbulence statistics change from the weak regime ($C_n^2 = 10^{-16}$) to moderately strong conditions ($C_n^2 = 10^{-15}$) at the beam center, an increase in SNR of about 27.7 dB is required to compensate for the effects of the change in the pdf of I_ν .

In order to get further insight on the outage probability, consider the no turbulence case (9) with $\sigma_j \neq 0$. The modified Bessel function can be expressed in a series as

$$I_0(x) = \sum_{k=0}^{\infty} \frac{((1/4)x^2)^k}{k! \Gamma(k+1)}. \quad (12)$$

Since all the terms are positive, we can get a lower bound by using only the first term in the series expansion. Thus, we get

$$p(I_s) \geq \left(\frac{\Omega}{\sigma_j}\right)^2 \exp\left(-\frac{A^2}{2\sigma_j^2}\right) I_s^{(\Omega^2/\sigma_j^2)-1}.$$

Then, the CDF bound is

$$F_s(I_s) \geq \exp\left(-\frac{A^2}{2\sigma_j^2}\right) I_s^{\Omega^2/\sigma_j^2}.$$

Thus, $P_{\text{out}} = F_s(I_{so})$ gives

$$I_{so} \leq P_{\text{out}}^{(\sigma_j^2/\Omega^2)} \exp\left(\frac{A^2}{2\sigma_j^2}\right). \quad (13)$$

$$p(I_s) = \begin{cases} \left(\frac{\Omega}{\sigma_j}\right)^2 \left(\frac{1}{I_s}\right) \exp\left(-\frac{1}{2\sigma_j^2}(A^2 - 2\Omega^2 \ln(I_s))\right) \\ \times I_0\left(\frac{A}{\sigma_j^2} \sqrt{2\Omega^2 \ln(1/I_s)}\right), & \sigma_j \neq 0 \\ \delta\left(I_s - \exp\left(-\frac{A^2}{2\Omega^2}\right)\right), & \sigma_j = 0 \end{cases} \quad (9)$$

The use of the approximate expression on the right-hand side of (13) will provide us an upper bound on capacity. This expression is quite accurate for low boresight and high jitter scenarios, and can also provide approximate results for weak turbulence conditions. Writing $\sigma_j^2/\Omega^2 = n$, we see that an increase in jitter rapidly decreases I_{so} at least as rapidly as P_{out}^n . This degrades capacity significantly.

D. Wave-Optics-Based Approach

We observe that the evaluation of the capacity expressions (8), (10), and (11) requires knowledge of parameter variations over the transverse plane. Since models of variations for $\alpha(\rho)$ and $\beta(\rho)$ with ρ are still under development, we propose here a wave-optics-based approach to evaluate the channel capacity. A wave (physical) optics simulation involves modeling the propagation of a 2-D spatially sampled optical field using scalar diffraction principles. Our simulation is developed using AO Tools [19], a toolbox for the MATLAB environment that includes components for modeling propagation through atmospheric turbulence. For one beam realization, the initial Gaussian beam receives a random tilt to simulate jitter, then the beam is propagated through a series of discrete random phase screens (turbulence screens) distributed between the source and the receiving plane. The turbulence screens assume a Kolmogorov spectrum with a supplemental component to model low-order tilt and the screen amplitudes encode the appropriate turbulence strength (C_n^2 value). The irradiance pattern at the receiving plane is found by computing the magnitude-squared value of the complex field.

For this study, the initial Gaussian beam size is 1 cm ($1/e^2$ irradiance radius) and a wavelength of $1 \mu\text{m}$ is assumed. Jitter is simulated by applying a random phase tilt (two-axis) to the initial beam field. We model a horizontal propagation path length of 10 km where the value of C_n^2 is kept constant over the path. We examine three different turbulence strength cases with C_n^2 values of 10^{-16} , 10^{-15} , and 10^{-14} . For our path length, these cases correspond to weak, medium, and strong fluctuation regimes, respectively. The corresponding Rytov variance values for an on axis point are 0.226, 2.26, and 22.6, respectively. The values of W_e are 0.321, 0.344, and 0.522 m, respectively. Three turbulence screens, evenly distributed along the path, are used for the $C_n^2 = 10^{-16}$ and $C_n^2 = 10^{-15}$ cases and five screens are used for the $C_n^2 = 10^{-14}$ case. In choosing the number of screens, we observed the criterion that the phase structure function for each screen be less than 1 rad^2 for adjacent samples in the screen mesh. The simulation mesh size is 256×256 points.

We generate 10 000 beam realizations for each scenario where the random seed for the turbulence screens and the random seed for the jitter phase tilt are changed for each realization. We record the resulting mean irradiance profile at the receiving plane and the series of 10 000 irradiance values for the point on the beam axis and 40 other points off the axis (corresponding to various boresight errors). The positions of these points and their index values are shown in Fig. 1. Neighboring points on either the horizontal or the vertical axis are effectively spaced 4 cm apart in the receiving plane.

The irradiance data are next used to fit turbulence pdf models at each point shown in Fig. 1. As expected, the pdf parameters are found to vary at the observation points. These variations

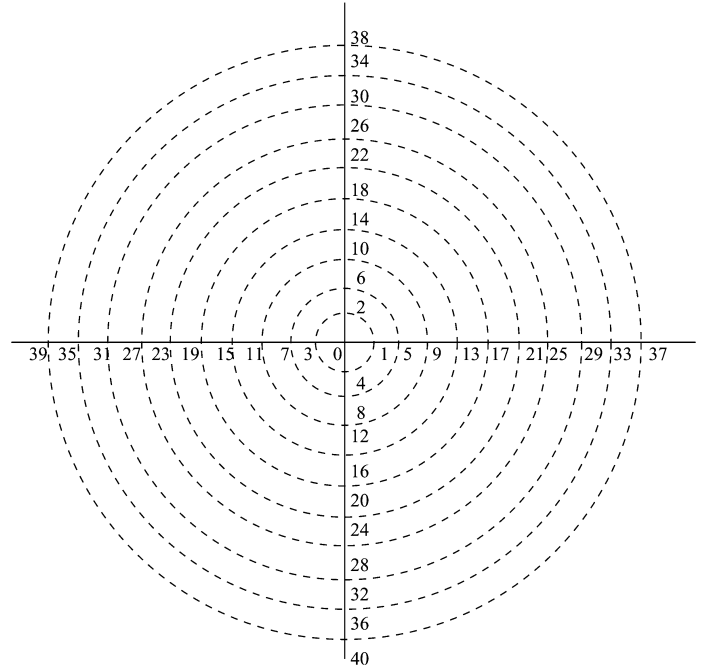


Fig. 1. Irradiance measurement locations on the receiver plane.

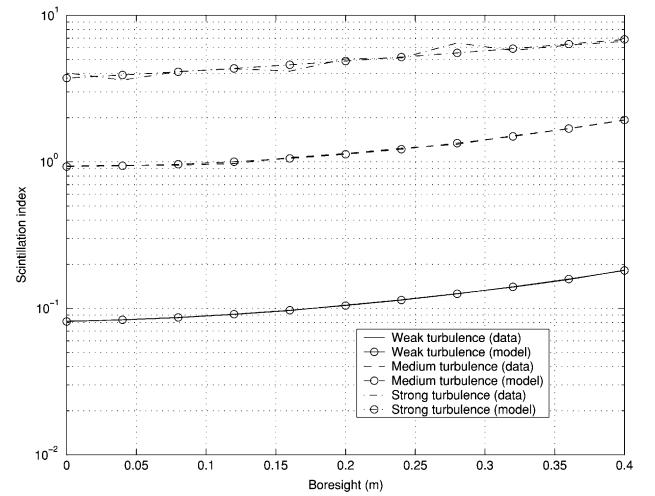


Fig. 2. Scintillation index variations with boresight.

are modeled using nonlinear functions. This relies on the assumption that the parameters σ , α , and β vary smoothly over the receiving plane. The parameter ψ is modeled as $\log \psi = a_\psi + b_\psi r + c_\psi r^2$, where ψ is σ in the LN model, and α , β , or the scintillation index σ_I^2 in the gamma-gamma model. The constants a_ψ , b_ψ , and c_ψ are estimated from the observed data. In our numerical results, we use the gamma-gamma model. When the gamma-gamma model is valid, the three parameters α , β and σ_I^2 are related as [13]

$$\sigma_I^2 = \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\alpha\beta}. \quad (14)$$

We show the variation of σ_I^2 against boresight in Fig. 2. Observe that our nonlinear model fits the observed data very well for the weak turbulence ($C_n^2 = 10^{-16}$) and the moderate turbulence ($C_n^2 = 10^{-15}$) conditions. In the strong turbulence

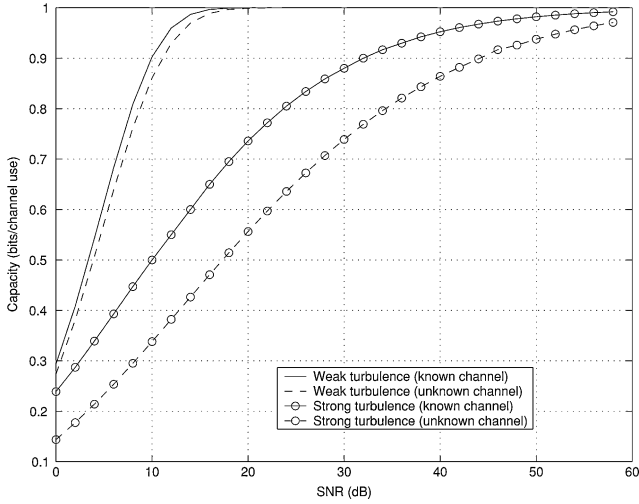


Fig. 3. Ergodic capacity variations with SNR. The boresight and jitter values are kept fixed at zero.

($C_n^2 = 10^{-14}$) case, there are slight variations from the model. In all cases, σ_I^2 at a boresight of 0.40 m is found to be almost double of the beam center value. For example, in the weak turbulence case, it increases from 0.082 at the beam center to a value of 0.182 at a boresight of 0.40 m. We also observed that fitting of the gamma-gamma model in the strong turbulence case is not as good as the weak and the moderate turbulence cases. However, minor variations in α and β did not affect our results in any significant way. Note that similar deviations from the gamma-gamma model for certain cases were also mentioned by the original authors in [20].

E. Discussion on Capacity Results

The ergodic capacity results for both known and unknown channels are shown in Fig. 3. The boresight and the jitter values are both set at zero. In the weak turbulence regime, a faster increase in capacity values with an increase in the SNR is observed. A similar behavior was observed for the unknown channel case in [4]. We also see that the gap between known and unknown channels is high for the stronger turbulence scenarios. When the turbulence is strong, fluctuations in the irradiance values increase, and thus, not knowing the channel state imposes a larger penalty on the channel capacity.

In Fig. 4, we show channel capacity versus various boresight values for the known channel case at an SNR of 15 dB in the absence of any jitter. We observe that the capacity decreases rapidly with an increase in boresight for no turbulence and weak turbulence cases. However, for strong turbulences, the capacity decreases very slowly with boresight. Thus, strong turbulence channels are robust to pointing error fluctuations. This comes at a price of lower capacity values when the pointing is accurate. However, when the pointing error is high, e.g., at $A = 0.34$ m, the medium turbulence case provides more than 40% gain over weak turbulences even after removing the effect of beam center irradiance normalization. It should be cautioned that very strong turbulences can eventually lower effective capacity gains. We also note that at zero boresight, the capacity difference for various turbulence channels is purely due to different statistics of the received signals. As the boresight increases, the capacity change is due to both loss in mean intensity as well as the change

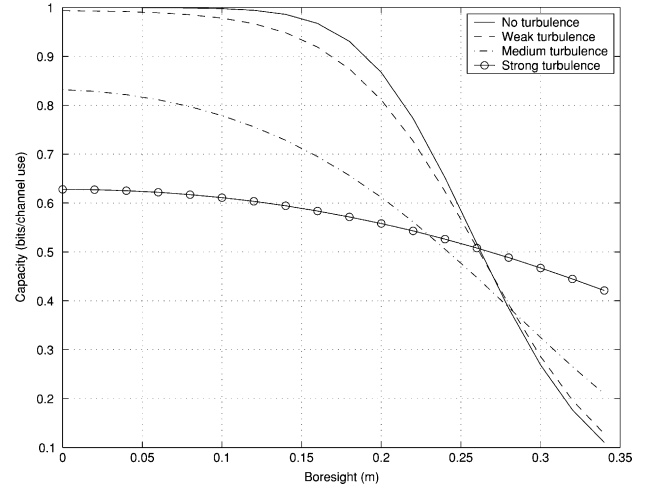


Fig. 4. Ergodic capacity variations with boresight at an SNR of 15 dB. The jitter is fixed at zero.

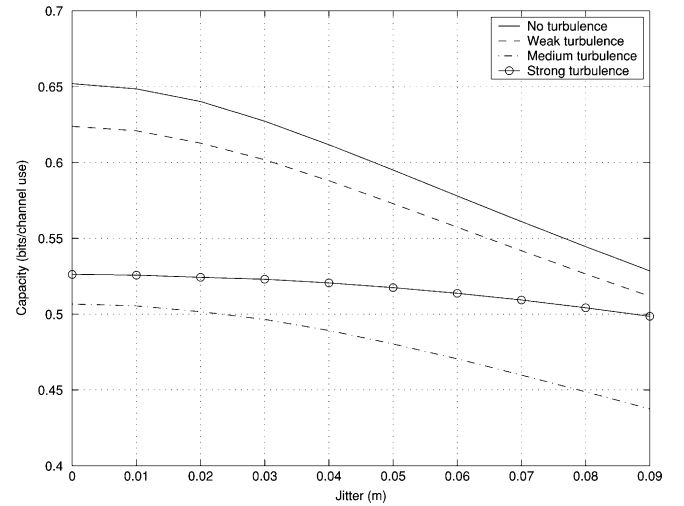


Fig. 5. Ergodic capacity variations against jitter values for a fixed boresight of 0.24 m. The SNR is 15 dB.

in the normalized irradiance pdfs. For the weak turbulence conditions, the mean intensity as well as the pdf change rapidly as we move away from the beam center.

Fig. 5 shows capacity variations with jitter values for a boresight of 0.24 m for the known channel case. Jitter variations are analogous to random variations in boresights. Unlike a fixed boresight error, jitter creates both better and worse pointing samples corresponding to movements toward or away from the mean beam center. The results can be interpreted by referring back to Fig. 4 for a 0.24-m boresight. As the jitter becomes higher, larger swings on both sides of the 0.24 m point produce reduction in capacity for the medium turbulence case, while it remains nearly same for the strong turbulence. Similarly, the no turbulence case has higher capacity than the weak turbulence case in Fig. 5. For large swings, this gap slightly reduces as the capacity for the weak turbulence becomes higher for larger boresight (Fig. 4), and the capacity gap is low at low boresight.

Another comment on the effect of jitter is that unlike boresight, an increase in jitter does not necessarily mean a decrease in the mean irradiance. Depending on the values of W_e and A ,

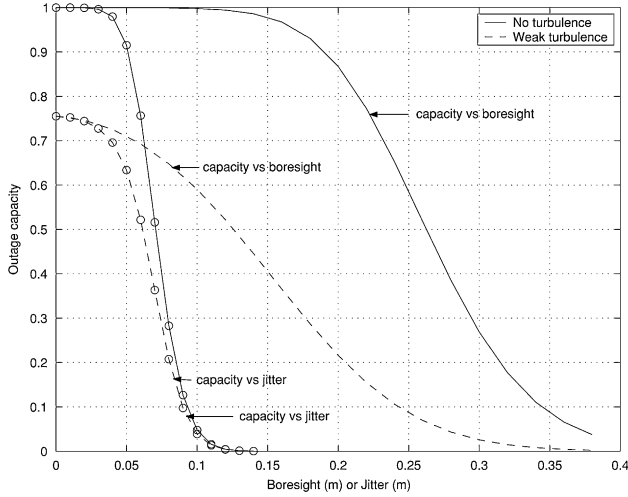


Fig. 6. Outage capacity for different boresight and jitter values at 15 dB SNR. The outage probability is 0.001.

an increase in σ_j can instead increase the mean intensity until a certain jitter value is reached. To find the best σ_j , we differentiate (5) with respect to σ_j^2 and equate it to zero to obtain

$$\sigma_{j,\text{best}}^2 = \frac{2A^2 - W_e^2}{4}. \quad (15)$$

Thus, when $A > W_e/\sqrt{2}$, a nonzero jitter increases the average received power and the best value is given by the previous expression.

Fig. 6 shows outage capacity for an outage probability $P_{\text{out}} = 10^{-3}$ at 15 dB SNR. Two sets of curves are shown: one shows outage capacity versus boresight while keeping jitter fixed at zero, while the other set is against jitter while keeping boresight fixed at zero. We observe that even weaker turbulence brings outage capacity significantly lower from the no turbulence capacity. This is unlike ergodic capacity where weak turbulence offers only a slight degradation over the no turbulence case. Note that the outage capacity is affected by $I_{so} = \gamma_s F_\nu^{-1}(P_{\text{out}})$, and in this case, although the value of γ_s does not change significantly from the no turbulence case to the weak turbulence case, the change of pdf affects $F_\nu^{-1}(P_{\text{out}})$ significantly.

Aperture averaging (AA) can reduce the impact of scintillations. The aperture averaged intensity can be approximately obtained from discrete spatial samples as $(1/N) \sum_{n=1}^N I_s[n]$, where $I_s[n]$ is the n th spatial sample taken uniformly over a 2-D aperture and N is the total number of samples taken over the aperture. The expected aperture averaged value is $(1/N) \sum_{n=1}^N E[I_s[n]]$. For small apertures, the variations of turbulence pdf model parameters within the aperture can be ignored. In this case, the expected aperture averaged value can be approximately expressed in a Gaussian form [12] as $K_A \exp(-(2r^2)/(W_A^2))$, where r is the distance of the circular aperture's center from the beam center, and K_A and W_A depend on the size of the aperture and W_e . In Fig. 7, we show the outage capacity results for moderately strong turbulence ($C_n^2 = 10^{-15}$) at a boresight value of $A = 0.16$ m. The AA2 and the AA4 detectors represent AA with apertures of diameter 2 and 4 cm, respectively. We also show capacity results for a fixed-parameter case, where the pdf model parameters are

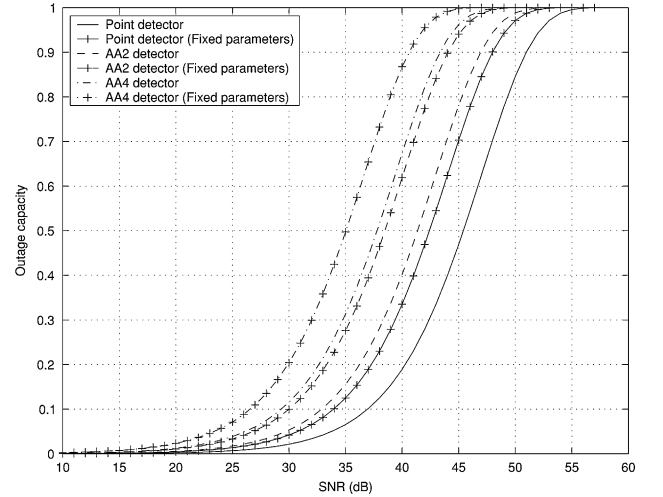


Fig. 7. Effects of AA on outage capacity at 15 dB SNR. The outage probability is 0.001.

unchanged from the values at the beam center. The fixed-parameter results, as considered in [12], thus provide an upper bound on capacity, as the scintillations tend to get stronger away from the beam center. AA can significantly increase capacity for stronger turbulence, as we see about 8 dB gain for the AA4 detector over the point detector in the middle region of the capacity curves.

V. ERROR PERFORMANCE ANALYSIS

In this section, we study the error performance of an FSO link affected by turbulence and pointing errors.

A. BER Results

We first present BER results for an uncoded FSO link in the presence of atmospheric turbulence and pointing errors. Both known and unknown channel cases under the ergodic scenario are considered. Note that the work in [1] considers only the unknown channel case without pointing errors. An ML symbol by symbol detector is used. First, we consider the known channel case. In a practical situation, the channel can be estimated using training symbols. Based on the received signal model (3), the receiver calculates the log-likelihood ratio (LLR) as

$$\text{LLR} = \ln \left(\frac{p(y|x=1)}{p(y|x=0)} \right) = \frac{I_s y}{\sigma_n^2} - \frac{I_s^2}{2\sigma_n^2}. \quad (16)$$

Thus, the decision threshold is $y_{\text{th}} = I_s/2$. The BER is

$$P_b = \text{Prob}(\text{error}|x=0)P(x=0) + \text{Prob}(\text{error}|x=1)P(x=1).$$

For equally likely symbols, the first term becomes

$$\frac{1}{2} \int_0^\infty Q \left(\frac{I_s}{2\sigma_n} \right) p(I_s) dI_s$$

where $Q(x) = 1/\sqrt{2\pi} \int_x^\infty \exp(-y^2/2) dy$ is the Gaussian tail probability. Finally, including the effects of pointing errors, the BER becomes

$$P_b = \int_0^\infty \int_0^\infty Q \left(\frac{I_s}{2\sigma_n} \right) p(I_s|\rho) p(\rho) dI_s d\rho. \quad (17)$$

This can be numerically evaluated. In the absence of atmospheric turbulence, using (9), P_b for $\sigma_j > 0$ becomes

$$P_b = \int_0^1 Q\left(\frac{I_s}{2\sigma_n}\right) \left(\frac{\Omega}{\sigma_j}\right)^2 \left(\frac{1}{I_s}\right) \exp(-1/(2\sigma_j^2)) \times (A^2 - 2\Omega^2 \ln(I_s)) I_0 \left(\frac{A}{\sigma_j^2} \sqrt{2\Omega^2 \ln(1/I_s)}\right) dI_s. \quad (18)$$

Note that the work [10] also presents BER expressions for the zero boresight case without considering parameter variations over the receiver plane.

When the channel state is unknown but the pdf of the output is known, the LLR can be written as

$$\text{LLR} = \ln \left(\frac{\int_0^\infty (1/\sqrt{2\pi\sigma_n^2}) \exp(-(y - I_s)^2/2\sigma_n^2) p(I_s) dI_s}{(1/\sqrt{2\pi\sigma_n^2}) \exp(-y^2/2\sigma_n^2)} \right) \quad (19)$$

$$= \ln \left(\int_0^\infty \exp\left(\frac{yI_s}{\sigma_n^2} - \frac{I_s^2}{2\sigma_n^2}\right) p(I_s) dI_s \right). \quad (20)$$

For a given pdf model $p(I_s)$, we can numerically calculate the threshold value y_{th} for which the LLR in (20) is zero. This is calculated only once. For equally likely symbols, the bit error probability is

$$P_b = 0.5Q\left(\frac{y_{th}}{\sigma_n}\right) + 0.5 \int_0^\infty \left(\left(1 - Q\left(\frac{y_{th} - I_s}{\sigma_n}\right)\right) \times \mathcal{I}_{\{y_{th} > I_s\}} + Q\left(\frac{I_s - y_{th}}{\sigma_n}\right) \mathcal{I}_{\{y_{th} \leq I_s\}} \right) p(I_s) dI_s \quad (21)$$

where $\mathcal{I}_{\{x\}}$ is an indicator function, which is 1 if x is true and is 0 otherwise.

In Fig. 8, the uncoded BER results for the known channel case are shown for the weak turbulence case for an SNR of 25 dB. We also show results for the fixed-parameter case, when the turbulence pdf model parameter variation over the receiver plane is ignored as in [10] and [11]. This fixed-parameter approach is found to provide a lower bound. Note that the difference in the uncoded error performance between the fixed-parameter approach and our approach can translate into large performance difference between the two approaches in coded systems. We also performed a study using channel codes, for example, a regular (3,6) low-density parity-check (LDPC) code [21]. Although the results are not shown here, the BER performance was found to degrade rapidly with boresight, caused mostly by the loss in average intensity.

B. Outage Probability Results

When the channel variations are not fast enough, the channel can remain fixed over the duration of one or more codewords. However, over the duration of a large number of codewords, the channel can change significantly. Since codes, such as the LDPC codes, show threshold effects, we can study outage probability corresponding to a threshold SNR to characterize the performance of such channels. Fig. 9 shows P_{out} for different boresight values at an SNR of 15 dB using a threshold SNR value of 4.5 dB for $C_n^2 = 10^{-16}$. In order to get more insight, we also show P_{out} with renormalized signal values. The renormalization is done by removing the effect of boresight on average signal, i.e., the average signal without jitter is set at unity for

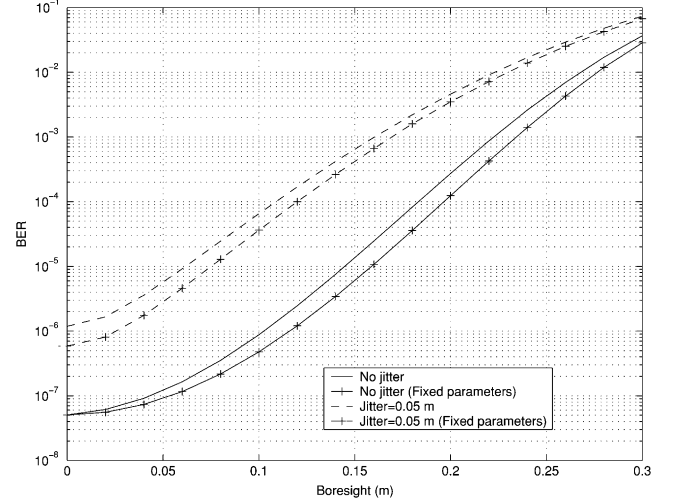


Fig. 8. BER analysis results for uncoded systems at 25 dB SNR.

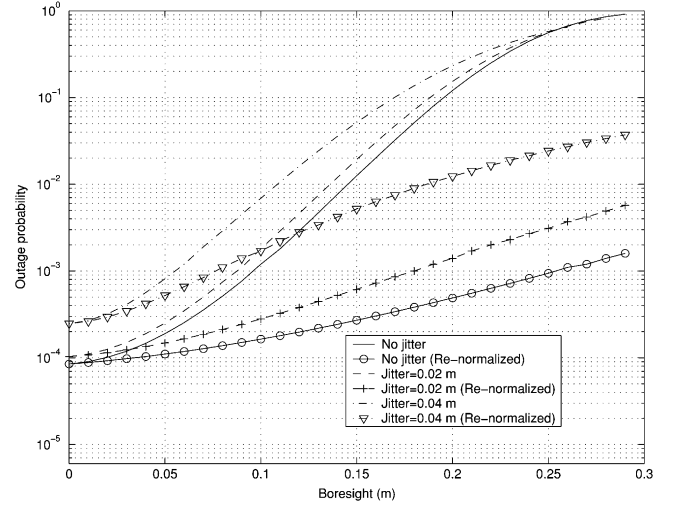


Fig. 9. Outage probability performance at 15 dB SNR.

each boresight value. In the no jitter case, the renormalization curve directly shows the effects of change of the pdf structure while eliminating the effects of average signal decrease due to boresight. Observe that for large boresight, the degradation in P_{out} due to changes in scintillation statistics becomes significant. Also, for high jitter, the change in the pdf structure plays an important role in the outage probability.

VI. CONCLUSION

We have considered the link performance of a laser system in the presence of pointing errors and atmospheric turbulence. An expression for the mean irradiance is provided. Ergodic channel capacity expressions for both known and unknown channel states are presented. The outage capacity equations are also given. A wave-optics-based approach is proposed to evaluate the capacities for various turbulence scenarios and pointing parameters. This method can be used by system designers in planning optical links with pointing uncertainties. We demonstrate that, for the ergodic scenario, stronger turbulence can help mitigate pointing error effects to some extent. However, when the pointing is accurate, stronger turbulence

degrades performance significantly. We have also derived BER expressions for uncoded systems and demonstrate effects of pointing errors on outage probabilities. For low jitter, most of the degradation in outage probability due to boresight comes from the loss in average signal.

ACKNOWLEDGMENT

The authors would like to thank Dr. X. Xiao for valuable assistance on the AO Tools.

REFERENCES

- [1] X. Zhu and J. M. Kahn, "Free-space optical communication through atmospheric turbulence channels," *IEEE Trans. Commun.*, vol. 50, no. 8, pp. 1293–1300, Aug. 2002.
- [2] X. Zhu and J. M. Kahn, "Performance bounds for coded free-space optical communications through atmospheric turbulence channels," *IEEE Trans. Commun.*, vol. 51, no. 8, pp. 1233–1239, Aug. 2003.
- [3] J. Li and M. Uysal, "Optical wireless communications: System model, capacity and coding," in *Proc. IEEE VTC Fall*, Orlando, FL, Oct. 2003, pp. 168–172.
- [4] J. Anguita, I. Djordjevic, M. A. Neifeld, and B. Vasic, "Shannon capacities and error-correction codes for the optical atmospheric channel," *OSA J. Opt. Netw.*, vol. 4, pp. 586–601, Aug. 2005.
- [5] D. K. Borah, D. G. Voelz, and S. Basu, "Maximum-likelihood estimation of a laser system pointing parameters by use of return photon counts," *Appl. Opt.*, vol. 45, pp. 2504–2509, Apr. 2006.
- [6] S. Arnon, "Power versus stabilization for laser satellite communication," *Appl. Opt.*, vol. 38, pp. 3229–3233, May 1999.
- [7] D. Kedar and S. Arnon, "Optical wireless communication through fog in the presence of pointing errors," *Appl. Opt.*, vol. 42, pp. 4946–4954, Aug. 2003.
- [8] V. V. Nikulin and R. Khandekar, "Performance of laser communication uplinks and downlinks in the presence of pointing errors and atmospheric distortions," *Proc. SPIE*, vol. 5712, pp. 37–45, 2005.
- [9] D. K. Borah and D. G. Voelz, "Estimation of laser beam pointing parameters in the presence of atmospheric turbulence," *Appl. Opt.*, vol. 46, pp. 6010–6018, Aug. 2007.
- [10] S. Arnon, "Effects of atmospheric turbulence and building sway on optical wireless-communication systems," *Opt. Lett.*, vol. 28, pp. 129–131, Jan. 2003.
- [11] H. G. Sandalidis, T. A. Tsiftsis, G. K. Karagiannidis, and M. Uysal, "BER performance of FSO links over strong atmospheric turbulence channels with pointing errors," *IEEE Commun. Lett.*, vol. 12, no. 1, pp. 44–46, Jan. 2008.
- [12] A. A. Farid and S. Hranilovic, "Outage capacity optimization for free-space optical links with pointing errors," *J. Lightw. Technol.*, vol. 25, no. 7, pp. 1702–1710, Jul. 2007.
- [13] L. C. Andrews, R. L. Phillips, and C. Y. Hopen, *Laser Beam Scintillation With Applications*. Bellingham, WA: SPIE, 2001.
- [14] I. S. Gradshteyn and I. M. Ryzhik, *Table of Integrals, Series, and Products*, 5th ed. New York: Academic, 1994.
- [15] J. H. Churnside and R. G. Frehlich, "Experimental evaluation of log-normal modulated Rician and IK models of optical scintillation in the atmosphere," *J. Opt. Soc. Amer.*, vol. A6, pp. 1760–1766, 1989.
- [16] C. C. Davis and I. I. Smolyaninov, "The effect of atmospheric turbulence on bit-error-rate in an on-off-keyed optical wireless system," *Proc. SPIE*, vol. 4489, pp. 126–136, 2002.
- [17] T. J. Richardson, M. A. Shokrollahi, and R. L. Urbanke, "Design of capacity-approaching irregular low-density parity-check codes," *IEEE Trans. Inf. Theory*, vol. 47, no. 2, pp. 619–637, Feb. 2001.
- [18] S.-Y. Chung, J. G. D. Forney, T. J. Richardson, and R. L. Urbanke, "On the design of low-density parity-check codes within 0.0045 db of the Shannon limit," *IEEE Commun. Lett.*, vol. 5, no. 2, pp. 58–60, Feb. 2001.
- [19] Adaptive Optics Toolbox [Online]. Available: http://www.mathworks.com/products/connections/product_main.shtml?prod_id=127 (n.d.) [Online], 2009.
- [20] M. A. Al-Habash, L. C. Andrews, and R. L. Phillips, "Mathematical model for the irradiance probability density function of a laser beam propagating through turbulent media," *Opt. Eng.*, vol. 40, pp. 1554–1562, Aug. 2001.
- [21] T. Richardson and R. Urbanke, *Modern Coding Theory*. Cambridge, U.K.: Cambridge Univ. Press, 2008.



Deva K. Borah (S'97–M'00) received the B.E. degree in electronics and communications engineering and the M.E. degree in electrical communication engineering from the Indian Institute of Science, Bengaluru, India, in 1987 and 1992, respectively, and the Ph.D. degree in telecommunications engineering from the Research School of Information Sciences and Engineering, Australian National University, Canberra, A.C.T., Australia, in 2000.

From 1988 to 1990, he was a Lecturer at Assam Engineering College, Guwahati, India. From 1992 to 1995, he was a Lecturer at Gauhati University, India. In Spring 2000, he joined the Klipsch School of Electrical and Computer Engineering, New Mexico State University, Las Cruces, where he is currently an Associate Professor. His current research interests include radio frequency and optical communications, digital signal processing, and statistical optics.



David G. Voelz received the B.S. degree in electrical engineering from New Mexico State University, Las Cruces, in 1981, and the M.S. and Ph.D. degrees in electrical engineering from the University of Illinois at Urbana-Champaign, Urbana, IL, in 1983 and 1987, respectively.

He is currently an Associate Professor of electrical engineering at New Mexico State University. From 1986 to 2001, he was with the Air Force Research Laboratory, Albuquerque, NM. His current research interests include free-space optical communications, spectral and polarimetric imaging, laser imaging, adaptive optics, and astronomical instrumentation development.